Flight to the Moon — Solution

Y26 (2026) — English translation

A1 — 0.50 pts

Write the expressions for $\ddot{y}$ through $\dot{y}$, k, m_0, μ, u, g and t.

Let us consider the change in the momentum of the system. Let at some moment the mass of the rocket be m and its speed $\dot{y}$. After a short time dt the mass of the rocket became less by μdt, and the speed of the rocket increased by $\ddot{y}dt$. In this case, the combustion products in the Earth's reference frame have a speed $u - \dot{y}$ directed downward. In this case, two external forces act on the system: air resistance and the gravity of the Earth, therefore the law of change in momentum

$$m\dot{y} - mgdt - k\dot{y}\,|\dot{y}|\,dt = (m - \mu dt)(\dot{y} + \ddot{y}dt) - \mu dt(u - \dot{y}).$$

Express $\ddot{y}$

$$-mg - k\dot{y}\,|\dot{y}| = m\ddot{y} - \mu u.$$

$$\ddot{y} = \frac{\mu u}{m} - g - \frac{k\dot{y}\,|\dot{y}|}{m}$$

It remains to express it through the required quantities.

Answer:

$$\ddot{y} = \frac{\mu u}{m_0 - \mu t} - g - \frac{k\dot{y}\,|\dot{y}|}{m_0 - \mu t}$$

A2 — 1.00 pts

Neglecting the drag force, find the speed of the rocket $v(m)$ and the coordinate $y(m)$ at the moment when its mass is m. Express your answer in terms of m, g, u, μ and m_0.

Let's use the answer from the previous paragraph and integrate twice, neglecting the resistance force, substituting $t = \dfrac{m_0 - m}{\mu}$

$$\ddot{y}dt = \frac{\mu u}{m}dt - gdt,$$

$$\int_0^v d\dot{y} = -\int_{m_0}^{m} u\frac{dm}{m} - \int_0^t gdt,$$

Integrating the expression for speed over time

$$\int_0^y dy = -u\int_0^t \ln\left(1 - \frac{\mu t}{m_0}\right)dt - \int_0^t gtdt.$$

For the integral of the logarithm it is known that $\int \ln x dx = x\ln x - x + C$

$$y = \frac{um_0}{\mu}\left(\left(1 - \frac{\mu t}{m_0}\right)\ln\left(1 - \frac{\mu t}{m_0}\right) - \left(1 - \frac{\mu t}{m_0}\right) + 1\right) - \frac{gt^2}{2},$$

Answer:

$$v = -u\ln\frac{m}{m_0} - g\frac{m_0 - m}{\mu}$$

A3 **0.80 pts**

Using the result of the previous paragraph and assuming that k weakly depends on height, find numerically at what height H_c the rocket will be at the moment when the drag force becomes equal to the force of gravity.

Let us write down the condition for the equality of gravity and the resistance force

$$kv^2 = mg,$$

$$k\left(-u\ln\frac{m}{m_0} - g\frac{m_0 - m}{\mu}\right)^2 = mg.$$

Solve numerically equation relatively m, obtain $m = 209$ thus. Substitute mass in formula for height lift y.

Answer:

$$H_c = 124 \text{ km}$$

A4 **0.70 pts**

Considering a vertical take-off, numerically estimate the ratio β_1 of the initial to the final mass of the rocket, at which the rocket develops a speed v_a (the speed of movement in a circular orbit of radius a).

For motion in an orbit with radius a we have $v_a = \sqrt{\frac{GM_e}{a}}$, we substitute the expression for the speed v from point A2, we get a numerical answer.

Answer:

$$\beta_1 = 8.03$$

B1 **0.30 pts**

Consider a point on the Earth-Moon straight line between the Earth and the Moon, at which the gravitational forces of the Earth and the Moon are equal in magnitude. Find the distance r_c from the Moon to this point. Express your answer in terms of r_m, M_e and M_m.

Equality of power at a distance r_c

$$\frac{GM_m}{r_c^2} = \frac{GM_e}{(r_m - r_c)^2}$$

Transform obtain

$$r_c = \frac{r_m}{\sqrt{\frac{M_e}{M_m}} + 1}$$

$$\frac{GM_m}{r_c^2} = \frac{GM_e}{(r_m - r_c)^2}$$

Transforming we get

$$r_c = \frac{r_m}{\sqrt{\frac{M_e}{M_m}} + 1}$$

Answer:

$$r_c = \frac{r_m}{1 + \sqrt{\frac{M_e}{M_m}}} = 3.86 \cdot 10^7 \text{ m}$$

B2 **0.50 pts**

Considering the ship in the CO of the Moon, write down the expression for the specific angular momentum L_0^m relative to the center of the Moon and the speed v_2 at the moment when the distance to the Moon is

r_c. *Express your answer using s, s', c, v_1, v_m and r_c.*

The vector $\vec{v_2}$ is obtained from the velocity vector $\vec{v_1}$ by subtracting the vector $\vec{v_m}$, from which a velocity triangle can be formed. Using the cosine theorem

$$v_2 = \sqrt{v_1^2 + v_m^2 - 2v_m v_1 \sin\alpha}.$$

Let us write the specific angular momentum by definition

$$\vec{L_0^m} = [\vec{r_c}, \vec{v_1} - \vec{v_m}] = [\vec{r_c}, \vec{v_1}] - [\vec{r_c}, \vec{v_m}].$$

Take magnitude/modulus expression

$$L_0^m = r_c\left(v_1 \sin(\theta + \alpha) - v_m \cos\theta\right).$$

Answer:

$$v_2 = \sqrt{v_1^2 + v_m^2 - 2v_m v_1 s'}$$

B3 1.20 pts

By writing the ZSE in the Moon's CO for the position of the ship closest to the Moon and the position immediately after entering the attraction zone, obtain and solve the quadratic equation and obtain the exact expression for v_1/v_m. Express your answer in terms of s, s', c, b/r_c and v_{II}^m/v_m. Note. There is no need to select a root sign at this point. Do not forget that within the framework of the model used, the interaction with the Earth in this area does not need to be taken into account.

Note. There is no need to select a root sign at this point. Do not forget that within the framework of the model used, the interaction with the Earth in this area does not need to be taken into account.

Let the speed of the ship at the point closest to the Moon be v. Then from the law of conservation of momentum we obtain:

$$vb = L_0^m.$$

$$v = \frac{r_c}{b}\left(v_1 s - v_m c\right)$$

Law conservation energy:

$$\frac{v^2}{2} - \frac{GM_m}{b} = \frac{v_2^2}{2} - \frac{GM_m}{r_c}.$$

$$\frac{r_c^2}{b^2}\left(v_1 s - v_m c\right)^2 - \left(v_{\mathrm{II}}^m\right)^2 = v_1^2 + v_m^2 - 2v_m v_1 s' - \left(v_{\mathrm{II}}^m\right)^2 \frac{b}{r_c}$$

For convenience denote $Z = v_1/v_m$, $x = b/r_c$, $y = v_{\mathrm{II}}^m/v_m$.

$$\frac{1}{x^2}\left(Zs - c\right)^2 - y^2 = Z^2 + 1 - 2Zs' - y^2 x$$

$$Z^2\left(s^2 - x^2\right) - 2Z\left(sc - x^2 s'\right) + c^2 - x^2 - y^2 x^2 (1 - x) = 0$$

$$Z = \frac{sc - x^2 s' \pm \sqrt{(sc - x^2 s')^2 - (s^2 - x^2)(c^2 - x^2 - y^2 x^2 (1 - x))}}{s^2 - x^2}$$

Answer:

$$\frac{v_1}{v_m} = \frac{sc - \left(\frac{b}{r_c}\right)^2 s' \pm \sqrt{\left(sc - \left(\frac{b}{r_c}\right)^2 s'\right)^2 - \left(s^2 - \left(\frac{b}{r_c}\right)^2\right)\left(c^2 - \left(\frac{b}{r_c}\right)^2 - \left(\frac{v_{II}^m}{v_m}\right)^2 \left(\frac{b}{r_c}\right)^2 \left(1 - \left(\frac{b}{r_c}\right)\right)\right)}}{s^2 - \left(\frac{b}{r_c}\right)^2}$$

B4 1.00 pts

Show that the expression for v_1 obtained in paragraph B4, when expanded to first order in powers of b/r_c, is reduced to the form:

$$v_1 \approx \frac{v_m c}{s} + B_0 \cdot \frac{b}{r_c}$$

Express B_0 through v_m, v_{II}^m, s, c and s'. Choosing the sign of the root, show that $B_0 > 0$.

$$v_1 \approx \frac{v_m c}{s} + B_0 \cdot \frac{b}{r_c}$$

Express B_0 in terms of v_m, v_{II}^m, s, c and s'. When choosing a root sign, show that $B_0 > 0$.
We will expand to first order in x.

$$Z = \frac{sc - x^2 s' \pm \sqrt{(sc - x^2 s')^2 - (s^2 - x^2)(c^2 - x^2 - y^2 x^2 (1-x))}}{s^2 - x^2}$$

Consider expression under root to second-order accuracy (it can give a contribution of first order, if under root no/not term zero and first order, that hold in given case):

$$(sc - x^2 s')^2 - (s^2 - x^2)(c^2 - x^2 - y^2 x^2(1-x)) \approx s^2c^2 - 2scs'x^2 - \left(s^2c^2 - x^2\left(c^2 + s^2 + y^2 s^2\right)\right) = = \left(c^2 + s^2 + y^2 s^2 - 2scs'\right)x^2.$$

Expression for Z with accuracy to first order:

$$Z = \frac{sc \pm x\sqrt{c^2 + s^2 + y^2 s^2 - 2scs'}}{s^2}.$$

$$v_1 = \frac{v_m c}{s} \pm \frac{v_m \sqrt{c^2 + s^2 + \left(\frac{v_{\mathrm{II}}^m}{v_m}\right)^2 s^2 - 2scs'}}{s^2} \cdot \frac{b}{r_c}$$

It is easy to notice that without the second term we get the speed at which the ship flies exactly to the Moon. The "+" sign corresponds to the desired flight around the Moon, and the "-" sign corresponds to the flight from the other side.

Thus, the final expression for v_1 is:

$$v_1 = \frac{v_m c}{s} + \frac{v_m \sqrt{c^2 + s^2 + \left(\frac{v_{\mathrm{II}}^m}{v_m}\right)^2 s^2 - 2scs'}}{s^2} \cdot \frac{b}{r_c}$$

Answer:

$$B_0 = \frac{v_m \sqrt{c^2 + s^2 + \left(\frac{v_{\mathrm{II}}^m}{v_m}\right)^2 s^2 - 2scs'}}{s^2}$$

B5 0.60 pts

Show that the expression for v_1, up to first order, is:

$$v_1 \approx \frac{v_m}{tg\,\theta} + A\alpha + B\frac{b}{r_c}$$

Express A and B through v_m, v_{II}^m, θ.

$$v_1 \approx \frac{v_m}{\mathrm{tg}\,\theta} + A\alpha + B\frac{b}{r_c}$$

Express A and B in terms of v_m, v_{II}^m, θ.

Note that in the answer of the previous paragraph it is possible to expand B_0 to zero order, since when taking into account the corrections we will obtain terms of a high order of smallness (at least of the second type $\alpha b/r_c$).

$$B_0 \approx \frac{v_m\sqrt{\cos^2 + \sin^2\theta + \left(\frac{v_{\rm II}^m}{v_m}\right)^2 \sin^2\theta}}{\sin^2\theta} = \frac{v_m\sqrt{1 + \left(\frac{v_{\rm II}^m}{v_m}\right)^2 \sin^2\theta}}{\sin^2\theta}$$

Expand $v_m c/s$.

$$\frac{v_m c}{s} = \frac{v_m\cos\theta}{\sin(\alpha+\theta)} \approx \frac{v_m\cos\theta}{\sin\theta + \alpha\cos\theta} \approx \frac{v_m}{\mathrm{tg}\theta}\left(1 - \frac{\alpha}{\mathrm{tg}\theta}\right)$$

$$v_1 = \frac{v_m}{\mathrm{tg}\theta} - \frac{v_m}{\mathrm{tg}^2\theta}\alpha + \frac{v_m\sqrt{1 + \left(\frac{v_{\rm II}^m}{v_m}\right)^2 \sin^2\theta}}{\sin^2\theta} \cdot \frac{b}{r_c}$$

Answer:

$$A = -\frac{v_m}{\mathrm{tg}^2\theta}$$

$$B = \frac{v_m\sqrt{1 + \left(\frac{v_{\rm II}^m}{v_m}\right)^2 \sin^2\theta}}{\sin^2\theta}$$

B6 — 0.60 pts

Let's return to the Earth's reference system to find an expression for the angle α between the speed v_1 and the straight line Earth - Moon. Find an expression for α between the speed v_1 and the straight line Earth - Moon up to first order in powers r_c/r_m and a/r_m. Express your answer using a, r_c, r_m, v_1, v_0 and θ. Note. Do not forget that within the framework of the model used, interaction with the Moon in this area does not need to be taken into account.

Find an expression for α between the speed v_1 and the straight line Earth - Moon up to first order in powers r_c/r_m and a/r_m. Express your answer using a, r_c, r_m, v_1, v_0 and θ.

Note. Do not forget that within the framework of the model used, interaction with the Moon in this area does not need to be taken into account.

Let's use the law of conservation of angular momentum relative to the Earth. Consider the projections $\vec{v}_1$ and the radius vector from the Earth $\vec{r}$:

$$\vec{v}_1 = \begin{pmatrix} v_1\cos\alpha \\ v_1\sin\alpha \end{pmatrix},$$

$$\vec{r} = \begin{pmatrix} r_m - r_c\cos\theta \\ r_c\sin\theta \end{pmatrix}.$$

Law conservation momentum:

$$v_0 a = v_1\sin\alpha(r_m - r_c\cos\theta) - v_1 r_c\cos\alpha\sin\theta.$$

$$\frac{v_0 a}{v_1\sqrt{r_m^2 - 2r_m r_c\cos\theta + r_c^2}} = \sin\left(\alpha - \mathrm{arctg}\left(\frac{r_c\sin\theta}{r_m - r_c\cos\theta}\right)\right)$$

$$\alpha = \mathrm{arctg}\left(\frac{r_c\sin\theta}{r_m - r_c\cos\theta}\right) + \arcsin\left(\frac{v_0 a}{v_1\sqrt{r_m^2 - 2r_m r_c\cos\theta + r_c^2}}\right) \approx \frac{r_c\sin\theta}{r_m} + \frac{v_0 a}{v_1 r_m}$$

Answer: With

B7 — 0.40 pts

Writing the ZSE in the FR of the Earth and neglecting the terms of order a/r_m and r_c/r_m, express the speed v_0 in terms of v_1 and v_{II}^e. Note. Do not forget that within the framework of the model used, interaction with the Moon in this area does not need to be taken into account.

Note. Do not forget that within the framework of the model used, interaction with the Moon in this area does not need to be taken into account.

Law of conservation of energy:

$$\frac{v_0^2}{2} - \frac{GM_e}{a} = \frac{v_1^2}{2}.$$

Note, that potential energy in right part equation not account for, since it have order a/r_m.

$$v_0^2 - (v_{\mathrm{II}}^e)^2 = v_1^2$$

Answer:

$$v_0 = \sqrt{(v_{\mathrm{II}}^e)^2 + v_1^2}$$

B8 — 1.00 pts

By writing the equation of a conic section, you get an equation from which you can find θ, through v_1, α, v_m, G, M_m, b and r_c.

Let us write down the equation of the canonical section $r = \dfrac{p}{1 + e\cos\varphi}$. Let us write this equation for points whose distances are r_c and b

$$b = \frac{p}{1+e},$$

$$r_c = \frac{p}{1 - e\cos\varphi}.$$

Express from equation eccentricity and equate obtain

$$\frac{p}{b} - 1 = \left(1 - \frac{p}{r_c}\right) \cdot \frac{1}{\cos\theta}.$$

Substitute p by formula

$$p = \frac{(L_0^m m)^2}{GM_m m^2} = \frac{\left(v_1 \sin(\alpha+\theta) - v_m \cos\theta\right)^2 r_c^2}{GM_m}$$

Answer:

$$\frac{\left(v_1 \sin(\alpha+\theta) - v_m \cos\theta\right)^2 r_c^2}{GM_m b} - 1 = \left(1 - \frac{\left(v_1 \sin(\alpha+\theta) - v_m \cos\theta\right)^2 r_c}{GM_m}\right) \cdot \frac{1}{\cos\theta}$$

B9 — 0.40 pts

Find β_2 - the ratio of the initial mass of the rocket to the final mass when accelerating from speed v_a - orbital speed when moving in a circular orbit of radius a to speed v_0. Express your answer using u, v_{II}^e and v_a. Calculate the numerical value β_2, take the quantities necessary for calculations from part A.

Calculate the numerical value of β_2, take the quantities necessary for calculations from part A.

Let's write the differential equation from point A1 without taking into account the force of attraction and resistance:

$$mdv + udm = 0,$$

$$\int_{v_a}^{v_{II}^e} \frac{dv}{u} = -\int_{m_0}^{m} \frac{dm}{m},$$

$$\frac{v_{II}^e - v_a}{u} = -\ln\frac{m}{m_0},$$

$$\beta_2 = \exp\left(\frac{v_{II}^e - v_a}{u}\right)$$

$$mdv + udm = 0,$$

$$\int_{v_a}^{v_{II}^e} \frac{dv}{u} = -\int_{m_0}^{m} \frac{dm}{m},$$

$$\frac{v_{II}^e - v_a}{u} = -\ln\frac{m}{m_0},$$

$$\beta_2 = \exp\left(\frac{v_{II}^e - v_a}{u}\right)$$

Answer:

$$\beta_2 = \exp\left(\frac{\sqrt{2}-1}{u}\cdot\sqrt{\frac{GM_e}{a}}\right) = 2.05$$

C1 — 0.50 pts

Counting $v_2 = 1.5$ km/s *(take the remaining necessary quantities from the previous parts of the problem), calculate numerically how many times the mass of the rocket will decrease during the landing process* β_3.

Let's write down the ESE for moving the rocket from the beginning of the moon's gravitational zone to the point at which we enter an orbit close to the moon

$$\frac{mv_2^2}{2} - \frac{GM_m m}{r_c} = \frac{mv^2}{2} - \frac{GM_m m}{b},$$

$$v = \sqrt{v_2^2 - 2GM_m\left(\frac{1}{r_c} - \frac{1}{b}\right)}.$$

Writing the Tsiolkovsky formula obtained in part B9 with the substitution of the sign of the gas exhaust velocity

$$\frac{v}{u} = \ln\frac{m_0}{m_3}$$

$$\frac{v}{u} = \ln\frac{m_0}{m_3}$$

Answer:

$$\beta_3 = \exp\left(\frac{\sqrt{v_2^2 - 2GM_m\left(\frac{1}{r_c} - \frac{1}{b}\right)}}{u}\right) = 1.83$$

C2 — 0.50 pts

Calculate numerically what initial mass m the rocket must have to deliver cargo weighing 40 *t to the Moon.*

Answer:

$$m = \beta_1\beta_2\beta_3 m_{\text{weight}} \approx 2300 \text{ thus}$$